# NPN MEDIUM POWER TRANSISTORS IN SOT89

## Features

- $BV_{CEO}$ > 45V, 60V & 80V
- $I_c$ = 1A Continuous Collector Current
- $I_{CM}$ = 1.5A Peak Pulse Current
- Low Saturation Voltage $V_{CE(sat)}$ < 500mV @ 0.5A
- Gain Groups 10 and 16
- Epitaxial Planar Die Construction
- Complementary PNP types: BCX51, 52, and 53
- **Totally Lead-Free & Fully RoHS compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **Qualified to AEC-Q101 Standards for High Reliability**

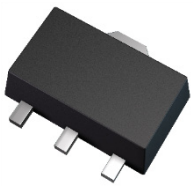
## Mechanical Data

- Case: SOT89
- Case Material: Molded Plastic, "Green" Molding Compound UL Flammability Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Matte Tin Finish Leads, Solderable per MIL-STD-202 Method 208 e3
- Weight: 0.055 grams (Approximate)

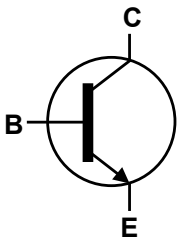
## Applications

- Medium Power Switching or Amplification Applications
- AF driver and output stages

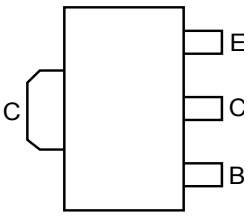
SOT89



Top View




Device Symbol



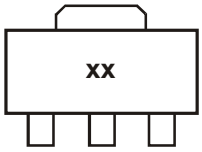

Top View
Pin-Out

## Ordering Information (Note 4)

| Product | Compliance | Marking | Reel size (inches) | Tape width (mm) | Quantity per reel |
|---|---|---|---|---|---|
| BCX54TA | AEC-Q101 | BA | 7 | 12 | 1,000 |
| BCX5410TA | AEC-Q101 | BC | 7 | 12 | 1,000 |
| BCX5416TA | AEC-Q101 | BD | 7 | 12 | 1,000 |
| BCX5416-13R | AEC-Q101 | BD | 13 | 12 | 4,000 |
| BCX55TA | AEC-Q101 | BE | 7 | 12 | 1,000 |
| BCX5510TA | AEC-Q101 | BG | 7 | 12 | 1,000 |
| BCX5516TA | AEC-Q101 | BM | 7 | 12 | 1,000 |
| BCX56TA | AEC-Q101 | BH | 7 | 12 | 1,000 |
| BCX5610TA | AEC-Q101 | BK | 7 | 12 | 1,000 |
| BCX5616TA | AEC-Q101 | BL | 7 | 12 | 1,000 |
| BCX5616TC | AEC-Q101 | BL | 13 | 12 | 4,000 |

Notes:
1. No purposely added lead. Fully EU Directive 2002/95/EC (RoHS) & 2011/65/EU (RoHS 2) compliant.
2. See http://www.diodes.com/quality/lead_free.html for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
4. For packaging details, go to our website at http://www.diodes.com/products/packages.html

## Marking Information




xx = Product Type Marking Code, as follows:

| | | |
|---|---|---|
| BCX54 = BA | BCX55 = BE | BCX56 = BH |
| BCX5410 = BC | BCX5510 = BG | BCX5610 = BK |
| BCX5416 = BD | BCX5516 = BM | BCX5616 = BL |


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## Absolute Maximum Ratings (@ $T_A$ = +25°C, unless otherwise specified.)

| Characteristic | Symbol | BCX54 | BCX55 | BCX56 | Unit |
|---|---|---|---|---|---|
| Collector-Base Voltage | $V_{CBO}$ | 45 | 60 | 100 | V |
| Collector-Emitter Voltage | $V_{CEO}$ | 45 | 60 | 80 | V |
| Emitter-Base Voltage | $V_{EBO}$ | 6 | | | V |
| Continuous Collector Current | $I_C$ | 1 | | | A |
| Peak Pulse Collector Current | $I_{CM}$ | 1.5 | | | |
| Continuous Base Current | $I_B$ | 100 | | | mA |
| Peak Pulse Base Current | $I_{BM}$ | 200 | | | |

## Thermal Characteristics (@$T_A$ = +25°C, unless otherwise specified.)

| Characteristic | | Symbol | Value | Unit |
|---|---|---|---|---|
| Power Dissipation | (Note 5) | $P_D$ | 1 | W |
| | (Note 6) | | 1.5 | |
| | (Note 7) | | 2.0 | |
| Thermal Resistance, Junction to Ambient Air | (Note 5) | $R_{\theta JA}$ | 125 | °C/W |
| | (Note 6) | | 83 | |
| | (Note 7) | | 60 | |
| Thermal Resistance, Junction to Lead | (Note 8) | $R_{\theta JL}$ | 13 | °C/W |
| Operating and Storage Temperature Range | | $T_J$, $T_{STG}$ | -65 to +150 | °C |

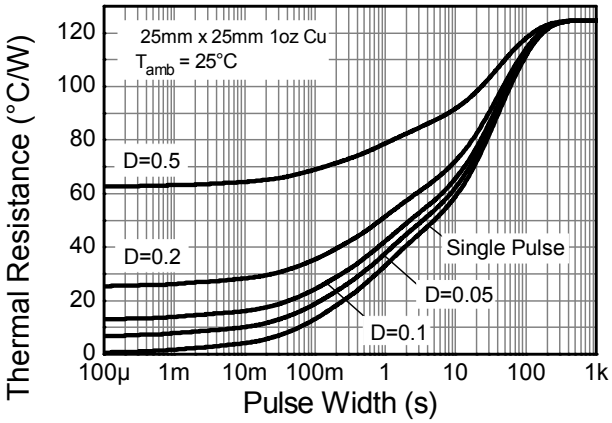
## ESD Ratings (Note 9)

| Characteristic | Symbol | Value | Unit | JEDEC Class |
|---|---|---|---|---|
| Electrostatic Discharge - Human Body Model | ESD HBM | 4,000 | V | 3A |
| Electrostatic Discharge - Machine Model | ESD MM | 400 | V | C |

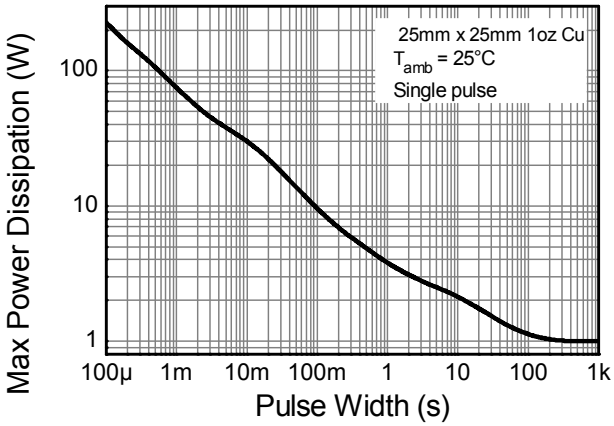
Notes:
5. For a device mounted with the exposed collector pad on 15mm x 15mm 1oz copper that is on a single-sided 1.6mm FR4 PCB; device is measured under still air conditions whilst operating in a steady-state.
6. Same as note (5), except the device is mounted on 25mm x 25mm 1oz copper.
7. Same as note (5), except the device is mounted on 50mm x 50mm 1oz copper.
8. Thermal resistance from junction to solder-point (on the exposed collector pad).
9. Refer to JEDEC specification JESD22-A114 and JESD22-A115.


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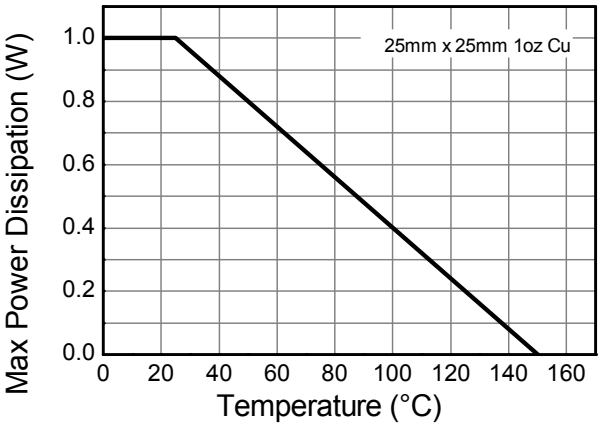
## Thermal Characteristics and Derating Information



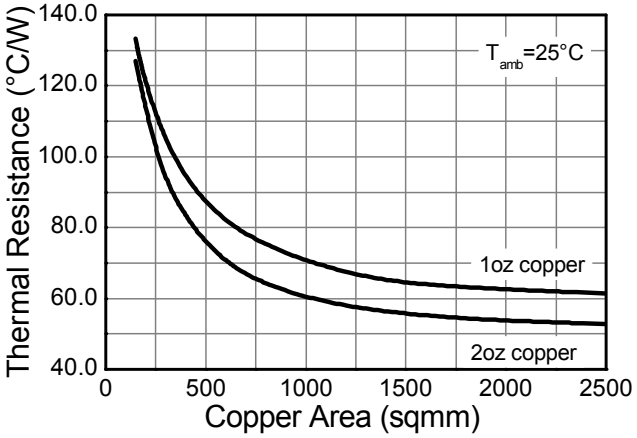

**Transient Thermal Impedance**

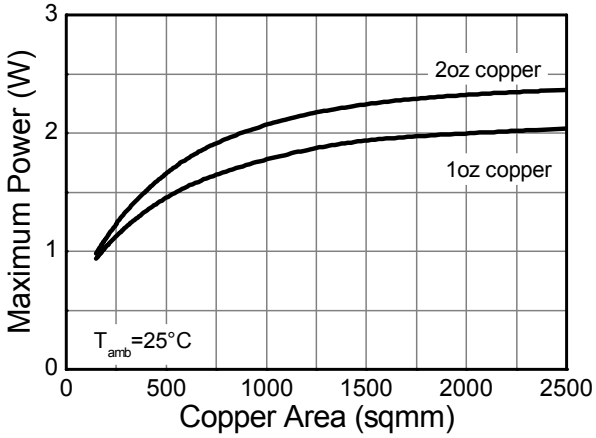



**Pulse Power Dissipation**




**Derating Curve**





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## Electrical Characteristics (@ $T_A$ = +25°C, unless otherwise specified.)

| Characteristic | | Symbol | Min | Typ | Max | Unit | Test Condition |
|---|---|---|---|---|---|---|---|
| Collector-Base Breakdown Voltage | BCX54 | $BV_{CBO}$ | 45 | — | — | V | $I_C$ = 100µA |
| | BCX55 | | 60 | | | | |
| | BCX56 | | 100 | | | | |
| Collector-Emitter Breakdown Voltage (Note 10) | BCX54 | $BV_{CEO}$ | 45 | — | — | V | $I_C$ = 10mA |
| | BCX55 | | 60 | | | | |
| | BCX56 | | 80 | | | | |
| Emitter-Base Breakdown Voltage | | $BV_{EBO}$ | 6 | — | — | V | $I_E$ = 100µA |
| Collector Cut-off Current | | $I_{CBO}$ | — | — | 0.1<br>20 | µA | $V_{CB}$ = 30V<br>$V_{CB}$ = 30V, $T_A$ = +150°C |
| Emitter Cut-off Current | | $I_{EBO}$ | — | — | 20 | nA | $V_{EB}$ = 5V |
| Static Forward Current Transfer Ratio (Note 10) | All versions | $h_{FE}$ | 25<br>40<br>25 | —<br>—<br>— | —<br>250<br>— | — | $I_C$ = 5mA, $V_{CE}$ = 2V<br>$I_C$ = 150mA, $V_{CE}$ = 2V<br>$I_C$ = 500mA, $V_{CE}$ = 2V |
| | 10 gain grp | | 63 | — | 160 | | $I_C$ = 150mA, $V_{CE}$ = 2V |
| | 16 gain grp | | 100 | — | 250 | | $I_C$ = 150mA, $V_{CE}$ = 2V |
| Collector-Emitter Saturation Voltage (Note 10) | | $V_{CE(sat)}$ | — | — | 0.5 | V | $I_C$ = 500mA, $I_B$ = 50mA |
| Base-Emitter Turn-On Voltage (Note 10) | | $V_{BE(on)}$ | — | — | 1.0 | V | $I_C$ = 500mA, $V_{CE}$ = 2V |
| Transition Frequency | | $f_T$ | 150 | — | — | MHz | $I_C$ = 50mA, $V_{CE}$ = 10V<br>f = 100MHz |
| Output Capacitance | | $C_{obo}$ | — | — | 25 | pF | $V_{CB}$ = 10V, f = 1MHz |

Note: 10. Measured under pulsed conditions. Pulse width ≤ 300µs. Duty cycle ≤ 2%.

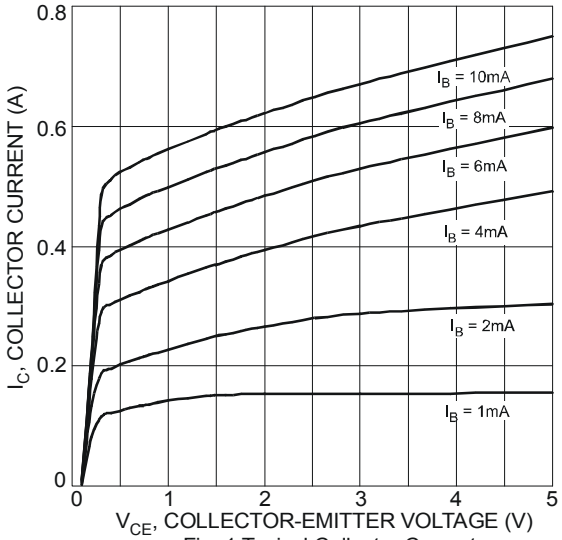



Fig. 1 Typical Collector Current vs. Collector-Emitter Voltage

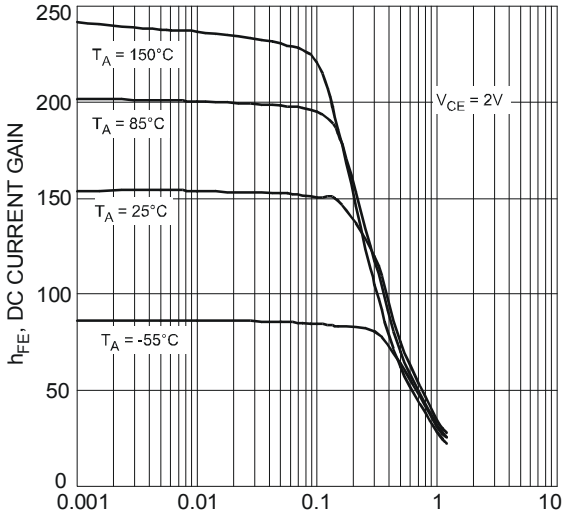



Fig. 2 Typical DC Current Gain vs. Collector Current


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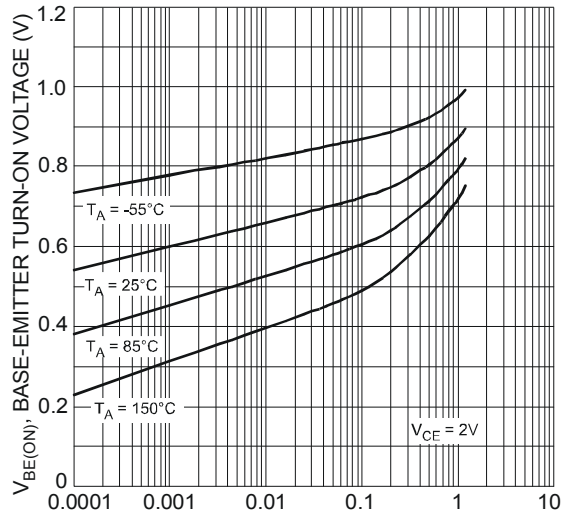



Fig. 3 Typical Base-Emitter Turn-On Voltage vs. Collector Current

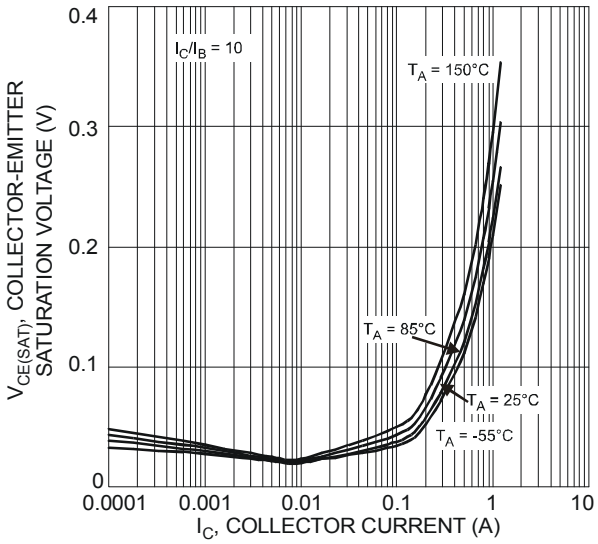



Fig. 4 Typical Collector-Emitter Saturation Voltage vs. Collector Current

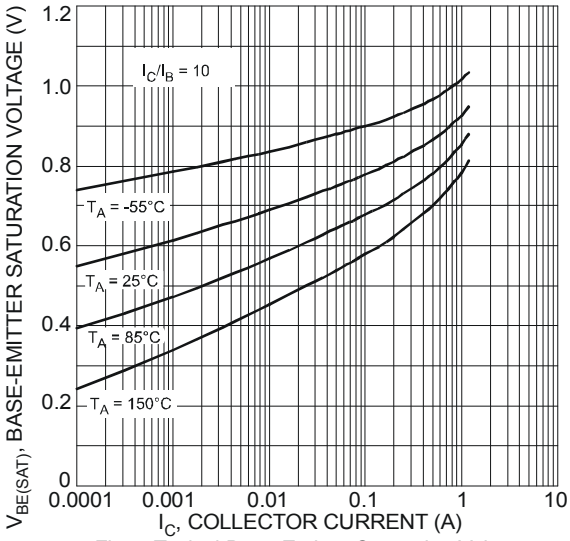



Fig. 5 Typical Base-Emitter Saturation Voltage vs. Collector Current

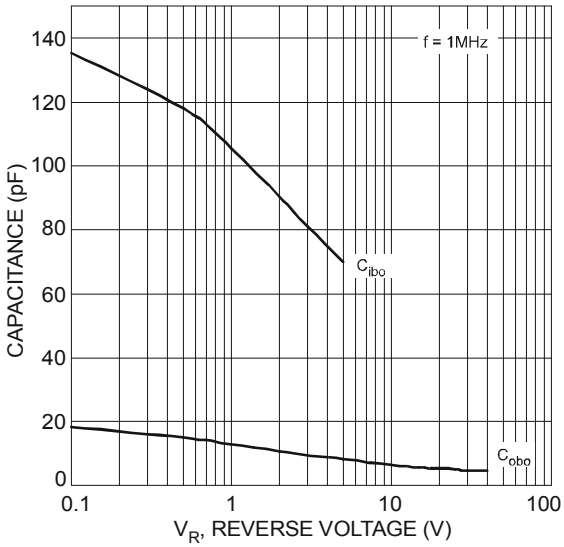



Fig. 6 Typical Capacitance Characteristics

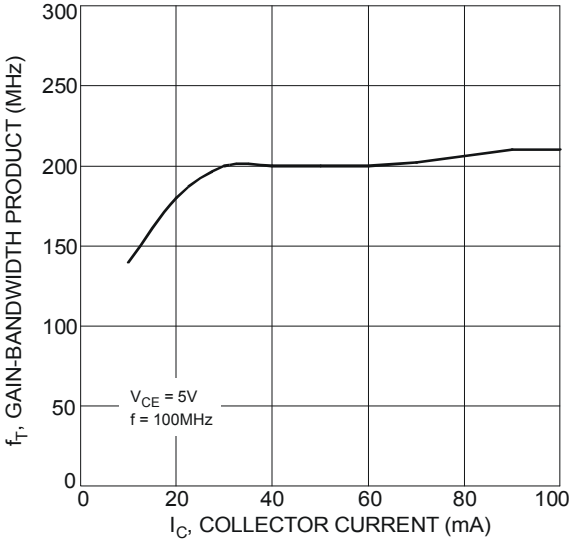
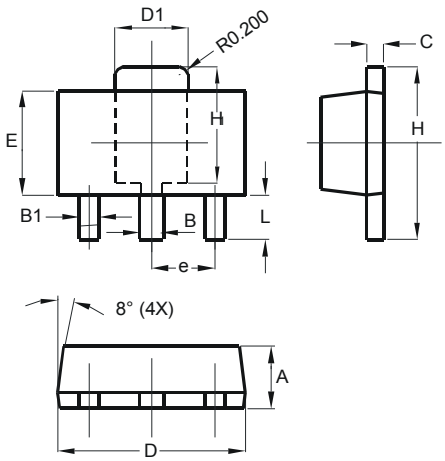



Fig. 7 Typical Gain-Bandwidth Product vs. Collector Current


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## Package Outline Dimensions

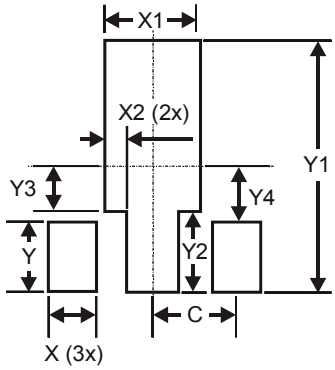
Please see AP02002 at http://www.diodes.com/datasheets/ap02002.pdf for latest version.




| SOT89 | | |
|---|---|---|
| **Dim** | **Min** | **Max** |
| **A** | 1.40 | 1.60 |
| **B** | 0.44 | 0.62 |
| **B1** | 0.35 | 0.54 |
| **C** | 0.35 | 0.44 |
| **D** | 4.40 | 4.60 |
| **D1** | 1.62 | 1.83 |
| **E** | 2.29 | 2.60 |
| **e** | 1.50 Typ | |
| **H** | 3.94 | 4.25 |
| **H1** | 2.63 | 2.93 |
| **L** | 0.89 | 1.20 |
| **All Dimensions in mm** | | |

## Suggested Pad Layout

Please see AP02001 at http://www.diodes.com/datasheets/ap02001.pdf for the latest version.




| Dimensions | Value (in mm) |
|---|---|
| **X** | 0.900 |
| **X1** | 1.733 |
| **X2** | 0.416 |
| **Y** | 1.300 |
| **Y1** | 4.600 |
| **Y2** | 1.475 |
| **Y3** | 0.950 |
| **Y4** | 1.125 |
| **C** | 1.500 |


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